Forum Math. **27** (2015), 1923–1944
DOI 10.1515/forum-2012-0167
Forum Mathematicum
© de Gruyter 2015

On the sum of two integral squares in certain quadratic fields

Dasheng Wei

Communicated by Jörg Brüdern

Abstract. In this note, we give a necessary and sufficient condition for determining which integers can be written as a sum of two integral squares for certain quadratic fields by using the integral Brauer–Manin obstruction. The condition is computable and originally from the reciprocity law.

Keywords. Integral points, ring class field, reciprocity law.

2010 Mathematics Subject Classification. 11E12, 11D09.

Introduction

It is significantly more difficult to study sums of two integral squares over algebraic number fields than that over $\mathbb{Z}$. From nowadays point of view, Fermat–Gauss' theorem about sums of two squares over $\mathbb{Z}$ is a purely local problem. However the question is a global problem over algebraic number fields, even quadratic fields since the class number of number fields is involved. There are only a few results about the question for general algebraic number fields F. Niven [7] studied the problem for $F = \mathbb{Q}(\sqrt{-1})$. This case is very special since the binary quadratic form x^2+y^2 is hyperbolic over $\mathbb{Q}(\sqrt{-1})$. Nagell [5,6] further studied the problem for $F = \mathbb{Q}(\sqrt{d})$ when d is one of the following twenty integers:

$$\pm 2,\ \pm 3,\ \pm 5,\ \pm 7,\ \pm 11,\ \pm 13,\ \pm 19,\ \pm 43,\ \pm 67,\ \pm 163.$$

His method follows Gauss' original idea and essentially depends on the fact that the class number of $\mathbb{Q}(\sqrt{d}, \sqrt{-d})$ is 1 when d is one of the above integers. Obviously this method cannot be applied to general algebraic number fields.

Recently Harari [2] proved that the integral Brauer–Manin obstruction (see [1] for definition) to the existence of an integral point is the only obstruction for an

The work is supported by the Morningside Center of Mathematics and grant DE 1646/2-1 of the Deutsche Forschungsgemeinschaft. The author is supported by NSFC, grant #10901150 and 973 Program 2013CB834202.

integral model of a principal homogenous space of tori. However, the Brauer group is infinite. One cannot use these results to determine the existence of integral points on a specific example. Fei Xu and the author gave another proof of the result in [11, 14]. In this paper we apply the method in [14] for the sum of two squares over quadratic fields.

It should be pointed out that the method in [14] only produces the idelic class groups of $F(\sqrt{-1})$ for solving the problem of a sum of two squares, where these idelic class groups are not unique. In other word, the finite subgroups of the Brauer group for testing the existence of the integral points are not unique. In order to get the explicit conditions for the sum of two squares, one needs further to construct the explicit abelian extensions of $F(\sqrt{-1})$ corresponding to the idelic class groups by class field theory. Such explicit construction is a wide open problem (Hilbert's 12th problem) in general but ad hoc methods exist.

Notation and terminology are standard if not explained. Let F be a number field, $\mathfrak{o}_F$ the ring of integers of F, Ω_F the set of all primes in F and ∞ the set of all infinite primes in F. For simplicity, we write $\mathfrak{p} < \infty$ for $\mathfrak{p} \in \Omega_F \setminus \infty$. Let $F_\mathfrak{p}$ be the completion of F at $\mathfrak{p}$ and $\mathfrak{o}_{F_\mathfrak{p}}$ be the local completion of $\mathfrak{o}_F$ at $\mathfrak{p}$ for each $\mathfrak{p} \in \Omega_F$. Write $\mathfrak{o}_{F_\mathfrak{p}} = F_\mathfrak{p}$ for $\mathfrak{p} \in \infty$. We also denote the adele ring (resp. the idele ring) of F by $\mathbb{A}_F$ (resp. $\mathbb{I}_F$).

Suppose that -1 is not a square in F. Let $E = F(\sqrt{-1})$. Let $\mathbf{X}_\alpha$ denote the affine scheme over $\mathfrak{o}_F$ defined by the equation $x^2 + y^2 = \alpha$ for a non-zero integer $\alpha \in \mathfrak{o}_F$. The equation $x^2 + y^2 = \alpha$ is solvable over $\mathfrak{o}_F$ if and only if $\mathbf{X}_\alpha(\mathfrak{o}_F) \neq \emptyset$. Let $X_\alpha = \mathbf{X}_\alpha \times_{\mathfrak{o}_F} F$. Obviously $f = x + y\sqrt{-1}$ is an invertible function on $X_\alpha \times_F E$, and f induces a natural map

$$f_E : X_\alpha(\mathbb{A}_F) \to \mathbb{I}_E.$$

The restriction to $X_\alpha(F_\mathfrak{p})$ of f_E can be defined by

$$f_E[(x_\mathfrak{p}, y_\mathfrak{p})] = \begin{cases} (x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, x_\mathfrak{p} - y_\mathfrak{p}\sqrt{-1}) \in E_{v_1} \oplus E_{v_2} & \text{if } \mathfrak{p} \text{ splits in } E/F, \\ x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1} \in E_v & \text{otherwise,} \end{cases}$$

where v_1 and v_2 (resp. v) are places of E above $\mathfrak{p}$.

Definition 0.1. Let $K_1, \ldots, K_n$ be finite abelian extensions over E. Let

$$\psi_{K_i/E} : \mathbb{I}_E \to \mathrm{Gal}(K_i/E) \quad \text{for } 1 \le i \le n$$

be the Artin map. We say that α satisfies the Artin condition of $K_1, \ldots, K_n$ if there is

$$\prod_{\mathfrak{p} \le \infty} (x_\mathfrak{p}, y_\mathfrak{p}) \in \prod_{\mathfrak{p} \le \infty} \mathbf{X}_\alpha(\mathfrak{o}_{F_\mathfrak{p}})$$

such that

$$\psi_{K_i/E}\Big(f_E\Big[\prod_{\mathfrak{p}\le\infty}(x_\mathfrak{p}, y_\mathfrak{p})\Big]\Big) = 1_i \quad \text{for } i = 1, \dots, n,$$

where 1_i is the identity element of $\mathrm{Gal}(K_i/E)$.

By class field theory, it is a necessary condition for $\mathbf{X}_\alpha(\mathfrak{o}_F) \neq \emptyset$ that α satisfies the Artin condition of $K_1, \dots, K_n$. There exists a (non-unique) finite abelian extension K/E independent of α, such that the Artin condition of K is also sufficient for $\mathbf{X}_\alpha(\mathfrak{o}_F) \neq \emptyset$ (see [14, Corollary 1.6, Theorem 1.10]). Let $d \geq 2$ be a positive square-free integer and $F = \mathbb{Q}(\sqrt{-d})$. Then the field K closely depends on the (local and global) solvability of the following equation:

$$x^2 + y^2 = -1. \tag{0.1}$$

Let $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$ and H_L be the ring class field corresponding to the order L. For example, the Artin condition of H_L is sufficient if $F = \mathbb{Q}(\sqrt{p})$ or $\mathbb{Q}(\sqrt{-p})$ with p prime [10, Theorem 0.3], and some related works were also performed in [12, 13].

Proposition 0.2 ([10, Proposition 1.1]). *Suppose one of the following conditions holds:*

(1) *The equation* (0.1) *has an integral solution in* $\mathfrak{o}_F$.

(2) *The equation* (0.1) *has no local integral solutions at a place of* F.

Then the equation $x^2 + y^2 = \alpha$ *is solvable over* $\mathfrak{o}_F$ *if and only if* α *satisfies the Artin condition of* H_L.

Therefore we only need to consider the case that the two conditions in Proposition 0.2 do not hold. For this case the Artin condition of H_L is not sufficient since -1 satisfies the Artin condition of H_L but the equation (0.1) is not solvable. In general the question will be very complicated. We cannot expect to construct a field K generally such that the Artin condition of K is sufficient.

The following result is well known [9, Satz 2]: the equation (0.1) is solvable over $\mathfrak{o}_F$ if and only if the equation

$$x^2 - dy^2 = -\gamma(d) \tag{0.2}$$

is solvable over $\mathbb{Z}$, where

$$\gamma(d) = \begin{cases} 1 & \text{if } d \not\equiv -1 \bmod 4, \\ 2 & \text{if } d \equiv -1 \bmod 4. \end{cases}$$

In this paper, we will consider the case that the equation (0.1) is solvable over $\mathfrak{o}_{F_\mathfrak{p}}$ for any $\mathfrak{p} \in \Omega_F$ and the equation (0.2) is not solvable over $\mathbb{Z}_p$ for some prime p. It is easy to see that the two conditions in Proposition 0.2 don't hold for this case. Let

$$\begin{aligned} C &= \{(d, p) \mid d \text{ is a square-free positive integer and } p|d \text{ with } p \text{ prime}\}, \\ D_1 &= \{(d, p) \in C \mid d \not\equiv -1 \bmod 8, p \equiv -1 \bmod 8\}, \\ D_2 &= \{(d, p) \in C \mid d \equiv 1, 2 \bmod 4, p \equiv 3 \bmod 8\}, \\ D_3 &= \{(d, p) \in C \mid d \equiv 3 \bmod 8, p \equiv 5 \bmod 8\}. \end{aligned}$$

Denote

$$D = D_1 \cup D_2 \cup D_3.$$

We say that $d \in D$ (or D_i) if there is a prime p such that $(d, p) \in D$ (or D_i). It is easy to verify that the equation (0.1) is solvable over $\mathfrak{o}_{F_\mathfrak{p}}$ for any $\mathfrak{p} \in \Omega_F$ and the equation (0.2) is not solvable over $\mathbb{Z}_p$ for some prime p if and only if $d \in D$. In this paper, we mainly prove the following result.

Theorem 0.3. *Let $(d, p) \in D$ and $F = \mathbb{Q}(\sqrt{-d})$. Then the Diophantine equation $x^2 + y^2 = \alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of Θ and H_L, where $\Theta = E(\sqrt[4]{p})$.*

Sums of two integral squares over real quadratic fields are more complicated than that over imaginary quadratic fields, since there are infinitely many units in real quadratic fields. In this paper, we also obtain a result about real quadratic fields.

Theorem 0.4. *Let p be a prime and $F = \mathbb{Q}(\sqrt{2p})$. If $p \equiv 3 \bmod 8$, then the Diophantine equation $x^2 + y^2 = \alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of Θ and H_L, where $\Theta = E(\sqrt[4]{2})$.*

As application, we explicitly determine which integers can be written as a sum of two integral squares for the two quadratic fields $\mathbb{Q}(\sqrt{\pm 6})$. Note that $\mathbb{Q}(\sqrt{\pm 6})$ are the fields with the smallest $|d|$ such that the Artin condition of H_L is not sufficient for $\mathbb{Q}(\sqrt{d})$.

1 The sum of two squares in imaginary quadratic fields

Let F be an algebraic number field such that -1 is not a square in F. Let $E = F(\sqrt{-1})$ and let T be the torus $R^1_{E/F}(\mathbb{G}_m) = \ker[R_{E/F}(\mathbb{G}_{m,E}) \to \mathbb{G}_{m,F}]$. Here

$R_{E/F}$ denotes Weil's restriction (see [4]). Denote by λ the embedding from T to $R_{E/F}(\mathbb{G}_{m,E})$. Obviously λ induces a natural group homomorphism

$$\lambda_E : T(\mathbb{A}_F) \to \mathbb{I}_E.$$

Let $\mathbf{X}_\alpha$ denote the affine scheme over $\mathfrak{o}_F$ defined by $x^2 + y^2 = \alpha$ for a non-zero integer $\alpha \in \mathfrak{o}_F$. Let $\mathbf{T}$ be the group scheme over $\mathfrak{o}_F$ defined by $x^2 + y^2 = 1$ and let $T = \mathbf{T} \times_{\mathfrak{o}_F} F$. The generic fiber of $\mathbf{X}_\alpha$ is a principal homogenous space of the torus T. Since $\mathbf{T}$ is separated over $\mathfrak{o}_F$, we can view $\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})$ as a subgroup of $T(F_\mathfrak{p})$.

Proposition 1.1 (cf. [14, Corollary 1.6]). *Let K_1/E and K_2/E be finite abelian extensions such that the group homomorphism $\tilde{\lambda}_E$ induced by λ_E,*

$$\tilde{\lambda}_E : {}^{T(\mathbb{A}_F)}/_{T(F)\prod_{\mathfrak{p}\le\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})} \to {}^{\mathbb{I}_E}/_{E^* N_{K_1/E}(\mathbb{I}_{K_1})} \times {}^{\mathbb{I}_E}/_{E^* N_{K_2/E}(\mathbb{I}_{K_2})},$$

is well-defined and injective, where well-defined means

$$\lambda_E\Big(T(F)\prod_{\mathfrak{p}\le\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})\Big) \subset (E^* N_{K_1/E}(\mathbb{I}_{K_1})) \cap (E^* N_{K_2/E}(\mathbb{I}_{K_2})).$$

Then $\mathbf{X}_\alpha(\mathfrak{o}_F) \neq \emptyset$ if and only if α satisfies the Artin condition of K_1 and K_2.

Let K be the composite field of K_1 and K_2, in fact the Artin condition of K is equivalent with that of K_1 and K_2. In this note, K_1 and K_2 have canonical definitions, hence we can compute the Artin condition of K by computing the Artin character of K_1/E and K_2/E respectively.

Let $d \geq 2$ be a square-free positive integer. Let $F = \mathbb{Q}(\sqrt{-d})$, $\mathfrak{o}_F$ be the ring of integers of F and $E = F(\sqrt{-1})$. In the rest of this section F is always an imaginary quadratic field. One takes the order $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$ inside E. Let H_L be the ring class field corresponding to the order $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$. Recall some notations:

$$\begin{aligned}
C &= \big\{(d,p) \mid d \text{ is a square-free positive integer and } p|d \text{ with } p \text{ prime}\big\},\\
D_1 &= \big\{(d,p) \in C \mid d \not\equiv -1 \bmod 8,\ p \equiv -1 \bmod 8\big\},\\
D_2 &= \big\{(d,p) \in C \mid d \equiv 1,2 \bmod 4,\ p \equiv 3 \bmod 8\big\},\\
D_3 &= \big\{(d,p) \in C \mid d \equiv 3 \bmod 8,\ p \equiv 5 \bmod 8\big\}.
\end{aligned}$$

Note that $p \equiv 3 \bmod 4$ if $(d,p) \in D_1 \cup D_2$.

Proposition 1.2. *Suppose* $(d, p) \in D_1 \cup D_2$ *and* $F = \mathbb{Q}(\sqrt{-d})$. *Let* $(x_\mathfrak{p}, y_\mathfrak{p}) \in \mathfrak{o}_{F_\mathfrak{p}} \times \mathfrak{o}_{F_\mathfrak{p}}$, *then the* 4*th Hilbert symbol*

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 = \begin{cases} 1 & \text{if } x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = 1, \\ -1 & \text{if } x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = -1, \end{cases}$$

where v *and* $\mathfrak{p}$ *are respectively the unique place of* E *and* F *above* p.

Proof. We have

$$\begin{aligned} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 &= \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, -d}{v}\Big)_4 \cdot \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, -d/p}{v}\Big)_4^{-1} \\ &= \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, -d}{v}\Big)_4 \cdot 1 \\ &= \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, \sqrt{-d}}{v}\Big) = \Big(\frac{\pm 1, p}{p}\Big) = \pm 1, \end{aligned}$$

where the second equation holds since $E(\sqrt[4]{-d/p})/E$ is unramified at v. □

Proposition 1.3. *Suppose* $(d, p) \in D_1 \cup D_2$ *and* $F = \mathbb{Q}(\sqrt{-d})$. *If* $x_\mathfrak{p}$ *and* $y_\mathfrak{p}$ *in* $\mathfrak{o}_{F_\mathfrak{p}}$ *satisfy* $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, *then*

$$\prod_{v|2} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 = 1,$$

where $v \in \Omega_E$ *and* $\mathfrak{p}$ *is the unique place of* F *above* 2.

The proposition follows from the following series of lemmas. Note that we do not assume that $d > 0$, l prime and $l|d$ in the following lemmas and some lemmas will be used in the real quadratic field case.

Lemma 1.4. *Let* $d \not\equiv -1 \bmod 8$ *and* $l \equiv -1 \bmod 8$. *Let* $F = \mathbb{Q}(\sqrt{-d})$. *If* $x_\mathfrak{p}$ *and* $y_\mathfrak{p}$ *in* $F_\mathfrak{p}$ *satisfy* $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, *then*

$$\prod_{v|2} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, l}{v}\Big)_4 = 1,$$

where $v \in \Omega_E$ *and* $\mathfrak{p}$ *is the unique place of* F *above* 2.

Proof. The extension $E(\sqrt{l})/E$ is split over v, hence

$$\sqrt{l} = s\sqrt{-1} = (2\sqrt{-1}) \cdot \frac{s}{2} = (1+\sqrt{-1})^2 \frac{2s}{2^2},$$

where $s^2 = -l$ and $s \in \mathbb{Z}_2^\times$. For any $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $F_\mathfrak{p}$ that satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, we have

$$\begin{aligned}\prod_{v|2}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, l}{v}\Big)_4 &= \prod_{v|2}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, \sqrt{l}}{v}\Big) \\ &= \prod_{v|2}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, 2s}{v}\Big) = \Big(\frac{\pm 1, 2s}{\mathfrak{p}}\Big) = 1. \qquad \square\end{aligned}$$

Proposition 1.3 for the case $(d, p) \in D_1$ follows from the above lemma. The following two lemmas deal with the case $(d, p) \in D_2$.

Lemma 1.5. *Let $d \equiv 1 \bmod 4$ and $l \equiv 3 \bmod 8$. Let $F = \mathbb{Q}(\sqrt{-d})$. If $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, then*

$$\prod_{v|2}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, l}{v}\Big)_4 = 1,$$

where $v \in \Omega_E$ and $\mathfrak{p}$ is the unique place of F above 2.

Proof. (1) If $d \equiv 5 \bmod 8$, then $F = \mathbb{Q}(\sqrt{-d}) = \mathbb{Q}(\sqrt{l})$ and v is the unique place of E above 2. For any $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $F_\mathfrak{p}$ that satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, we have

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, l}{v}\Big)_4 = \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, \sqrt{l}}{v}\Big) = \Big(\frac{\pm 1, \sqrt{l}}{\mathfrak{p}}\Big) = \Big(\frac{\pm 1, -l}{2}\Big) = 1.$$

(2) If $d \equiv 1 \bmod 8$, then E/F is split over $\mathfrak{p}$. For any $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ that satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, we have

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, l}{v}\Big) = 1$$

since $E(\sqrt{l})/E$ is unramified at any place v of E above 2. So we have

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, l}{v}\Big)_4 = \pm 1.$$

Therefore

$$\prod_{v|2}\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v}\Big)_4=\Big(\frac{(x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1})(x_\mathfrak{p}-y_\mathfrak{p}\sqrt{-1}),l}{\mathfrak{p}}\Big)_4$$
$$=\begin{cases}1 & \text{if } x_\mathfrak{p}^2+y_\mathfrak{p}^2=1,\\ \big(\frac{-1,l}{\mathfrak{p}}\big)_4=\big(\frac{\sqrt{-1},l}{\mathfrak{p}}\big)=1 & \text{if } x_\mathfrak{p}^2+y_\mathfrak{p}^2=-1.\end{cases}\qquad\square$$

Lemma 1.6. *Let $d\equiv 2 \bmod 4$ and $l\equiv\pm 3 \bmod 8$. Let $F=\mathbb{Q}(\sqrt{-d})$. If $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_\mathfrak{p}^2+y_\mathfrak{p}^2=\pm 1$, then*

$$\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v}\Big)_4=1,$$

where v and $\mathfrak{p}$ are respectively the unique place of E and F above 2.

Proof. (1) First we will prove that $(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v})_4=1$ for any $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ that satisfy $x_\mathfrak{p}^2+y_\mathfrak{p}^2=1$.

By Hilbert 90, there exists $\beta\in E_v^*$ such that $x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1}=\sigma(\beta)/\beta$, where σ is the non-trivial element of $\mathrm{Gal}(E_v/F_\mathfrak{p})$. Let $\beta=a+b\sqrt{-1}$ and we can choose β such that $a,b\in\mathfrak{o}_{F_\mathfrak{p}}$ and a or b is a unit. Denote $\kappa=a^2+b^2$.

Assume that $\mathrm{ord}_\mathfrak{p}(\kappa)$ is odd. The equation $x^2+y^2=\kappa$ is not solvable over $\mathfrak{o}_{F_\mathfrak{p}}$ if $\mathrm{ord}_\mathfrak{p}(\kappa)=1$ (by [3, Theorem 1]). Therefore $\mathrm{ord}_\mathfrak{p}(\kappa)\geq 3$. We have $a,b\in\mathfrak{o}_{F_p}^\times$ and

$$\beta/\sigma(\beta)=(a^2-b^2-2ab\sqrt{-1})\kappa^{-1}=1-2\kappa^{-1}b^2-2\kappa^{-1}ab\sqrt{-1}.$$

Since $\mathrm{ord}_\mathfrak{p}(\kappa)\geq 3$ and $a,b\in\mathfrak{o}_{F_\mathfrak{p}}^\times$, one has

$$2\kappa^{-1}b^2,2\kappa^{-1}ab\notin\mathfrak{o}_{F_\mathfrak{p}}.$$

So $\beta/\sigma(\beta)\notin\mathfrak{o}_{F_\mathfrak{p}}+\mathfrak{o}_{F_\mathfrak{p}}\sqrt{-1}$. A contradiction is derived. Thus, $\mathrm{ord}_\mathfrak{p}(\kappa)$ is even.

Let $\mathrm{ord}_\mathfrak{p}(\kappa)=2n$. Since E/F is totally ramified and of degree 2, one can write $\beta=\pi_F^n\mu$, where π_F is a uniformizer of F and $\mu\in\mathfrak{o}_{E_v}^\times$. Then $\sigma(\beta)/\beta=\sigma(\mu)/\mu$. Let $\mathfrak{q}$ be the unique place of $\mathbb{Q}(\sqrt{-1})$ above 2. We have

$$\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v}\Big)_4=\Big(\frac{\sigma(\mu)/\mu,l}{v}\Big)_4=\Big(\frac{N_{E_v/F_\mathfrak{p}}(\mu),l}{v}\Big)_4\cdot\Big(\frac{\mu,l}{v}\Big)^{-1}$$
$$=\Big(\frac{N_{E_v/\mathbb{Q}_2}(\mu),l}{\mathfrak{q}}\Big)_4\cdot\Big(\frac{N_{E_v/\mathbb{Q}_2}(\mu),l}{2}\Big)^{-1}.$$

Since $\mu \in \mathfrak{o}_{E_v}^{\times}$ and $E_v = \mathbb{Q}_2(\sqrt{-1}, \sqrt{-d})$ with $d \equiv 2 \bmod 4$, one has

$$N_{E_v/\mathbb{Q}_2}(\mu) = m^2 \quad \text{for some } m \in \mathbb{Z}_2^{\times}.$$

So

$$\Big(\frac{x_{\mathfrak{p}} + y_{\mathfrak{p}}\sqrt{-1}, l}{v}\Big)_4 = \Big(\frac{m^2, l}{\mathfrak{q}}\Big)_4 \cdot 1 = \Big(\frac{m, l}{\mathfrak{q}}\Big) = 1.$$

(2) Let $d = 2d_0$. Then d_0 is odd. By the argument in (1), we only need to show that there exists a tuple $(x_{\mathfrak{p}}, y_{\mathfrak{p}}) \in \mathfrak{o}_{F_{\mathfrak{p}}} \times \mathfrak{o}_{F_{\mathfrak{p}}}$ satisfying $x_{\mathfrak{p}}^2 + y_{\mathfrak{p}}^2 = -1$, such that

$$\Big(\frac{x_{\mathfrak{p}} + y_{\mathfrak{p}}\sqrt{-1}, l}{v}\Big)_4 = 1.$$

(i) If $d_0 \equiv 1 \bmod 4$, then $x^2 - 2d_0y^2 = -1$ is solvable over $\mathbb{Z}_2$. Choose one solution $(x_0, y_0) \in \mathbb{Z}_2 \times \mathbb{Z}_2$, then we have $x_0^2 + (y_0\sqrt{-2d_0})^2 = -1$. Let

$$x_{\mathfrak{p}} = x_0 \quad \text{and} \quad y_{\mathfrak{p}} = y_0\sqrt{-2d_0}.$$

One obtains

$$\Big(\frac{x_{\mathfrak{p}} + y_{\mathfrak{p}}\sqrt{-1}, l}{v}\Big)_4 = \Big(\frac{x_0 - y_0\sqrt{2d_0}, l}{v}\Big)_4 = \Big(\frac{-1, l}{\mathfrak{q}}\Big)_4 = \Big(\frac{\sqrt{-1}, l}{\mathfrak{q}}\Big) = 1.$$

Let $s \in \mathbb{Z}_2^{\times}$ such that $s^2 = l/3$ or $-l/3$. For any $\delta \in E_v^*$ that satisfies $N_{E_v/F_{\mathfrak{p}}}(\delta) = \pm 1$, we have

$$\Big(\frac{\delta, l}{v}\Big)_4 = \Big(\frac{\delta, 3}{v}\Big)_4 \cdot \Big(\frac{\delta, s}{v}\Big) \cdot \Big(\frac{\delta, \pm 1}{v}\Big)_4 = \Big(\frac{\delta, 3}{v}\Big)_4 \cdot \Big(\frac{\pm 1, s}{\mathfrak{p}}\Big) \cdot 1 = \Big(\frac{\delta, 3}{v}\Big)_4.$$

(ii) If $d_0 \equiv 3 \bmod 8$, then $F_{\mathfrak{p}} = \mathbb{Q}_2(\sqrt{-6})$. Let

$$x_{\mathfrak{p}} = (1 + 2\sqrt{-6})/5 \quad \text{and} \quad y_{\mathfrak{p}} = (2 - \sqrt{-6})/5.$$

Then $x_{\mathfrak{p}}^2 + y_{\mathfrak{p}}^2 = -1$. One has

$$\begin{aligned}\Big(\frac{x_{\mathfrak{p}} + y_{\mathfrak{p}}\sqrt{-1}, l}{v}\Big)_4 &= \Big(\frac{5^{-1}(1 + 2\sqrt{-1}) + 5^{-1}(2 - \sqrt{-1})\sqrt{-6}, 3}{v}\Big)_4 \\ &= \Big(\frac{5^{-1}(3 - 4\sqrt{-1}), 3}{\mathfrak{q}}\Big)_4 = \Big(\frac{3 - 4\sqrt{-1}, 3}{\mathfrak{q}}\Big)_4 \cdot \Big(\frac{5, 3}{\mathfrak{q}}\Big)_4^{-1}.\end{aligned}$$

It is easy to see that

$$\Big(\frac{5,3}{\mathfrak{q}}\Big)_4 = \Big(\frac{-3,3}{\mathfrak{q}}\Big)_4 = 1.$$

By class field theory, we have

$$\begin{aligned}\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v}\Big)_4 &= \prod_{w|3}\Big(\frac{3-4\sqrt{-1},3}{w}\Big)_4 \prod_{w|5}\Big(\frac{3-4\sqrt{-1},3}{w}\Big)_4\\ &= (-\sqrt{-1})^{(3^2-1)/4}\cdot\Big(\frac{5^2,3}{5}\Big)_4 = (-1)\cdot(-1) = 1,\end{aligned}$$

where w is in the set of places of $\mathbb{Q}(\sqrt{-1})$.

(iii) If $d_0 \equiv 7 \bmod 8$, then $F_\mathfrak{p} = \mathbb{Q}_2(\sqrt{-14})$. Let

$$x_\mathfrak{p} = (3-2\sqrt{-14})/13 \quad \text{and} \quad y_\mathfrak{p} = (2+3\sqrt{-14})/13.$$

Then $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = -1$. One has

$$\begin{aligned}\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v}\Big)_4 &= \Big(\frac{13^{-1}(3+2\sqrt{-1})+13^{-1}(-2+3\sqrt{-1})\sqrt{-14},3}{v}\Big)_4\\ &= \Big(\frac{13^{-1}(-5-12\sqrt{-1}),3}{\mathfrak{q}}\Big)_4\\ &= \Big(\frac{-5-12\sqrt{-1},3}{\mathfrak{q}}\Big)_4\cdot\Big(\frac{13,3}{\mathfrak{q}}\Big)_4^{-1}.\end{aligned}$$

It is easy to see that

$$\Big(\frac{13,3}{\mathfrak{q}}\Big)_4 = \Big(\frac{-3,3}{\mathfrak{q}}\Big)_4 = 1.$$

By class field theory, we have

$$\begin{aligned}\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},l}{v}\Big)_4 &= \prod_{w|3}\Big(\frac{-5-12\sqrt{-1},3}{w}\Big)_4 \prod_{w|13}\Big(\frac{-5-12\sqrt{-1},3}{w}\Big)_4\\ &= 1\cdot\Big(\frac{13^2,3}{13}\Big)_4 = 1\cdot 1 = 1,\end{aligned}$$

where w is in the set of places of $\mathbb{Q}(\sqrt{-1})$. □

Recall

$$D_3 = \big\{(d,p) \in C \mid d \equiv 3 \bmod 8,\ p \equiv 5 \bmod 8\big\}.$$

In the following, we give some properties about the case $(d,p) \in D_3$.

Proposition 1.7. *Let $(d, p) \in D_3$ and $F = \mathbb{Q}(\sqrt{-d})$. If $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \pm 1$, then*

$$\prod_{v|p} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v} \Big)_4 = 1,$$

where $v \in \Omega_E$ and $\mathfrak{p}$ is the unique place of F above p.

Proof. We have

$$\begin{aligned}
&\prod_{v|p} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v} \Big)_4 \\
&\quad = \prod_{v|p} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, -d}{v} \Big)_4 \cdot \prod_{v|p} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, -d/p}{v} \Big)_4^{-1} \\
&\quad = \prod_{v|p} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, -d}{v} \Big)_4 \cdot 1 = \prod_{v|p} \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, \sqrt{-d}}{v} \Big) \\
&\quad = \Big(\frac{\pm 1, p}{p} \Big) = 1,
\end{aligned}$$

where the second equation holds since $E(\sqrt[4]{-d/p})/E$ is unramified over v. □

Proposition 1.8. *Let $(d, p) \in D_3$ and $F = \mathbb{Q}(\sqrt{-d})$. Let $(x_\mathfrak{p}, y_\mathfrak{p}) \in \mathfrak{o}_{F_\mathfrak{p}} \times \mathfrak{o}_{F_\mathfrak{p}}$, then*

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v} \Big)_4 = \begin{cases} 1 & \text{if } x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = 1, \\ -1 & \text{if } x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = -1, \end{cases}$$

where v and $\mathfrak{p}$ are respectively the unique place of E and F above 2.

Proof. We have

$$\begin{aligned}
\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v} \Big)_4 &= \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, \sqrt{p}}{v} \Big) \\
&= \Big(\frac{\pm 1, \sqrt{p}}{\mathfrak{p}} \Big) = \Big(\frac{\pm 1, -p}{2} \Big) = \pm 1,
\end{aligned}$$

where the second equation holds since $F_\mathfrak{p} = \mathbb{Q}_2(\sqrt{-d}) = \mathbb{Q}_2(\sqrt{p})$. □

Now we can prove Theorem 0.3 by using the above propositions.

Proof of Theorem 0.3. Let $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$. Let $\mathfrak{p}$ be a place of F and $L_\mathfrak{p}$ be the $\mathfrak{p}$-adic completion of L inside $E_\mathfrak{p} = E \otimes_F F_\mathfrak{p}$. Recall $T = R^1_{E/F}(\mathbb{G}_{m,E})$ and $\mathbf{T}$ is the affine scheme defined by the equation $x^2 + y^2 = 1$. We have

$$T(F) = \big\{\beta \in E^* \mid N_{E/F}(\beta) = 1\big\},$$
$$\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}}) = \big\{\beta \in L_\mathfrak{p}^\times \mid N_{E_\mathfrak{p}/F_\mathfrak{p}}(\beta) = 1\big\},$$

and $L_\infty^\times = E_\infty^* = \mathbb{C}^* \times \mathbb{C}^*$.

Let $v \in \Omega_E$ and $v|2$. Let $\mathfrak{p}$ be the unique place of F above 2. By Propositions 1.3 and 1.8, one has

$$\prod_{v|2}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 = 1$$

for any $(x_\mathfrak{p}, y_\mathfrak{p}) \in \mathfrak{o}_{F_\mathfrak{p}} \times \mathfrak{o}_{F_\mathfrak{p}}$ with $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = 1$. Regard $\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})$ as a subgroup of $T(\mathbb{A}_F)$. This implies that

$$\lambda_E(\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})) \subseteq E^* N_{\Theta/E}(\mathbb{I}_\Theta).$$

Let $v \in \Omega_E$ and $v|p$. Let $\mathfrak{p}$ be the unique place of F above p. By Propositions 1.2 and 1.7, one has

$$\prod_{v|p}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 = 1$$

for any $(x_\mathfrak{p}, y_\mathfrak{p}) \in \mathfrak{o}_{F_\mathfrak{p}} \times \mathfrak{o}_{F_\mathfrak{p}}$ with $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = 1$. This implies that

$$\lambda_E(\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})) \subseteq E^* N_{\Theta/E}(\mathbb{I}_\Theta).$$

The field extension Θ/E is unramified over each place v of E except $v|2p$. Therefore the natural group homomorphism

$$\tilde{\lambda}_E : T(\mathbb{A}_F)/T(F)\prod_{\mathfrak{p}\le\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}}) \to \big[\mathbb{I}_E/E^* N_{\Theta/E}(\mathbb{I}_\Theta)\big] \times \big[\mathbb{I}_E/E^*\textstyle\prod_{\mathfrak{p}\le\infty} L_\mathfrak{p}^\times\big]$$

is well-defined. By Proposition 1.1, we only need to show $\tilde{\lambda}_E$ is injective.

Let $u \in \ker\tilde{\lambda}_E$. Then there are $\beta \in E^*$ and $i \in \prod_{\mathfrak{p}\le\infty} L_\mathfrak{p}^\times$ with $\lambda_E(u) = \beta i$. We have

$$N_{E/F}(\beta) = N_{E/F}(i)^{-1} \in F^* \cap \Big(\prod_{\mathfrak{p}\le\infty}\mathfrak{o}_{F_\mathfrak{p}}^\times\Big) = \{\pm 1\},$$

since F is an imaginary quadratic field and $F \neq \mathbb{Q}(\sqrt{-1})$ or $\mathbb{Q}(\sqrt{-3})$.

If $N_{E/F}(\beta) \neq 1$, one obtains $N_{E/F}(\beta) = N_{E/F}(i) = -1$. Write $i = (i_v)_v \in \mathbb{I}_E$. Since Θ/E is unramified over each place v of E except $v|2p$, one concludes

that $\psi_{\Theta/E}(i_v)$ is trivial for all primes $v \nmid 2p$, where i_v is regarded as an idele whose v-component is i_v and 1 otherwise.

(1) Suppose $(d, p) \in D_1 \cup D_2$. Since $N_{E/F}(i_v) = -1$ and $i_v \in L_{\mathfrak{p}}^{\times}$, one gets

$$\prod_{v|2} \psi_{\Theta/E}(i_v) = 1 \quad \text{and} \quad \psi_{\Theta/E}(i_{v'}) = -1$$

by Propositions 1.2 and 1.3, where $\psi_{\Theta/E} : \mathbb{I}_E \to \mathrm{Gal}(\Theta/E)$ is the Artin map and v' is the unique place of E above p. So

$$\psi_{\Theta/E}(\beta i) = \psi_{\Theta/E}(i) = \prod_{v|2} \psi_{\Theta/E}(i_v) \cdot \psi_{\Theta/E}(i_{v'}) = -1.$$

This contradicts to $u \in \ker \tilde{\lambda}_E$.

(2) Suppose $(d, p) \in D_3$. Since $N_{E/F}(i_v) = -1$ and $i_v \in L_{\mathfrak{p}}^{\times}$, one gets

$$\psi_{\Theta/E}(i_{v'}) = -1 \quad \text{and} \quad \prod_{v|p} \psi_{\Theta/E}(i_v) = 1$$

by Propositions 1.7 and 1.8, where v' is the unique place of E above 2. So

$$\psi_{\Theta/E}(\beta i) = \psi_{\Theta/E}(i) = \psi_{\Theta/E}(i_{v'}) \cdot \prod_{v|p} \psi_{\Theta/E}(i_v) = -1.$$

This contradicts to $u \in \ker \tilde{\lambda}_E$.

Therefore $N_{E/F}(\beta) = 1$, one concludes that

$$N_{E/F}(\beta) = N_{E/F}(i) = 1 \implies \beta \in T(F) \text{ and } i \in \prod_{\mathfrak{p} \le \infty} \mathbf{T}(\mathfrak{o}_{F_{\mathfrak{p}}}).$$

So $\beta i \in T(F) \prod_{\mathfrak{p} \le \infty} \mathbf{T}(\mathfrak{o}_{F_{\mathfrak{p}}})$. Then $\tilde{\lambda}_E$ is injective. □

Lemma 1.9. *Let $F = \mathbb{Q}(\sqrt{-2d})$ and $d \equiv 3 \bmod 4$. If $x_{\mathfrak{p}}$ and $y_{\mathfrak{p}}$ in $\mathfrak{o}_{F_{\mathfrak{p}}}$ satisfy $x_{\mathfrak{p}}^2 + y_{\mathfrak{p}}^2 = -1$, then for the 4th Hilbert symbol it holds*

$$\left(\frac{x_{\mathfrak{p}} + y_{\mathfrak{p}}\sqrt{-1}, 2}{v}\right)_4 = -1,$$

where v and $\mathfrak{p}$ are respectively the unique place of E and F above 2.

Proof. We have

$$\begin{aligned}\left(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},2}{v}\right)_4 &= \left(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},-2d}{v}\right)_4\cdot\left(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},-d}{v}\right)_4^{-1}\\ &= \left(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},-2d}{v}\right)_4\cdot 1 = \left(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},\sqrt{-2d}}{v}\right)\\ &= \left(\frac{-1,2d}{2}\right) = -1,\end{aligned}$$

where the second equation holds by Lemma 1.6. □

Using a similar argument as in the proof of Theorem 0.3, the next proposition follows from Lemmas 1.6 and 1.9.

Proposition 1.10. *Let $F=\mathbb{Q}(\sqrt{-2d})$ and $d\equiv 3 \bmod 4$. Then the Diophantine equation $x^2+y^2=\alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of Θ and H_L, where $\Theta=E(\sqrt[4]{2})$ and H_L is the ring class field corresponding to the order $L=\mathfrak{o}_F+\mathfrak{o}_F\sqrt{-1}$.*

Now we use Proposition 1.10 to give an explicit example.

Example 1.11. Let $F=\mathbb{Q}(\sqrt{-6})$. We write $N_{F/\mathbb{Q}}(\alpha)=2^{s_1}3^{s_2}p_1^{e_1}\cdots p_g^{e_g}$ for any $\alpha=a+b\sqrt{-6}$ and $a,b\in\mathbb{Z}$. Let $P(\alpha)=\{p_1,\dots,p_g\}$. Denote

$$\begin{aligned}P_1 &= \left\{p\in P(\alpha)\mid \left(\tfrac{-1}{p}\right)=\left(\tfrac{-6}{p}\right)=1 \text{ and } \left(\tfrac{2}{p}\right)=-1\right\},\\ P_2 &= \left\{p\in P(\alpha)\mid \left(\tfrac{-1}{p}\right)=-\left(\tfrac{-6}{p}\right)=1 \text{ and } \left(\tfrac{2}{p}\right)=-1\right\},\\ P_3 &= \left\{p\in P(\alpha)\mid \left(\tfrac{-1}{p}\right)=\left(\tfrac{-6}{p}\right)=1 \text{ and } \left(\tfrac{2}{p}\right)_4=-1\right\}.\end{aligned}$$

It is easy to see that e_i is even for $p_i\in P_2$.

Then $x^2+y^2=\alpha$ is solvable over $\mathfrak{o}_F$ if and only if

(1) The equation $x^2+y^2=\alpha$ has integral solutions at every place of F.

(2) $P_1\neq\emptyset$, or $2|a$, or

$$\sum_{p_i\in P_2} e_i/2+\sum_{p_i\in P_3} e_i\equiv\begin{cases}0 \bmod 2 & \text{if } a\equiv 1,3 \bmod 8,\\ 1 \bmod 2 & \text{if } a\equiv -1,-3 \bmod 8\end{cases}$$

for $P_1=\emptyset$ and $2\nmid a$.

2 The sum of two squares in real quadratic fields

Let $d > 1$ be a square-free odd number and $F = \mathbb{Q}(\sqrt{2d})$. Let $\mathfrak{o}_F$ be the ring of integers of F, ε_F the fundamental unit of $\mathfrak{o}_F$ and $\varepsilon_F = a + b\sqrt{2d}$ with $a, b > 0$. Let $E = F(\sqrt{-1})$. One takes the order $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$ inside E. In this section we always assume that one of the equations $x^2 - 2dy^2 = \pm 2$ is solvable over $\mathbb{Z}$ and we fix one solution (x_0, y_0). Denote $\omega = x_0 + y_0\sqrt{2d}$ and $\eta = \omega^2/2$. Then $\eta \in \mathfrak{o}_F^\times$ and $\eta = \varepsilon_F^{i_0}$ for some $i_0 \in \mathbb{Z}$. By the assumption, we have $N_{F/\mathbb{Q}}(\varepsilon_F) = 1$ (see [8, pp. 106–109]).

Lemma 2.1. *Let $p \equiv \pm 3 \bmod 8$ and $p|d$. If one of the equations $x^2 - 2dy^2 = \pm 2$ is solvable over $\mathbb{Z}$ with the notation as above, then i_0 is odd and*

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 = -1$$

for $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfying $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \varepsilon_F$, where v and $\mathfrak{p}$ are respectively the unique prime in E and F above p.

Proof. We have

$$\begin{aligned}\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 &= \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, 2d}{v}\Big)_4 \cdot \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, 2d/p}{v}\Big)_4^{-1} \\ &= \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, 2d}{v}\Big)_4 \cdot 1 = \Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, \sqrt{2d}}{v}\Big) \\ &= \Big(\frac{\varepsilon_F, \sqrt{2d}}{\mathfrak{p}}\Big),\end{aligned}$$

where the second equation holds since $E(\sqrt[4]{2d/p})/E$ is unramified over v. However,

$$\Big(\frac{\eta, \sqrt{2d}}{\mathfrak{p}}\Big) = \Big(\frac{\omega^2/2, \sqrt{2d}}{\mathfrak{p}}\Big) = \Big(\frac{2, p}{p}\Big) = -1.$$

Then

$$-1 = \Big(\frac{\eta, \sqrt{2d}}{\mathfrak{p}}\Big) = \Big(\frac{\varepsilon_F, \sqrt{2d}}{\mathfrak{p}}\Big)^{i_0}.$$

Therefore $\big(\frac{\varepsilon_F, \sqrt{2d}}{\mathfrak{p}}\big) = -1$ and i_0 is odd. □

Lemma 2.2. *Let $p|d$, $p \equiv \pm 3 \bmod 8$. Suppose one of the equations $x^2 - 2dy^2 = \pm 2$ is solvable over $\mathbb{Z}$ with the notation as above. If $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \varepsilon_F$, then*

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\Big)_4 = 1,$$

where v and $\mathfrak{p}$ are respectively the unique place of E and F above 2.

Proof. Since $E_v = F_\mathfrak{p}(\sqrt{-1})$ is ramified over $F_\mathfrak{p}$, there is a uniformizer π_F in $F_\mathfrak{p}$ such that $\big(\frac{\pi_F, -1}{\mathfrak{p}}\big) = 1$. We know $x^2 + y^2 = \pi_F^3$ is solvable over $\mathfrak{o}_{F_\mathfrak{p}}$ and $x^2 + y^2 = \pi_F$ is not solvable over $\mathfrak{o}_{F_\mathfrak{p}}$ (by [3, Theorem 1]). Choose one solution (a, b) of the equation $x^2 + y^2 = \pi_F^3$. Then we have a and $b \in \mathfrak{o}_{F_\mathfrak{p}}^\times$, otherwise we obtain that $x^2 + y^2 = \pi_F$ is solvable over $\mathfrak{o}_{F_\mathfrak{p}}$.

Recall $\omega = x_0 + y_0\sqrt{2d}$ and $\eta = \omega^2/2$, here $x_0, y_0 \in \mathbb{Z}$ satisfy $x_0^2 - 2dy_0^2 = \pm 2$. Denote

$$\epsilon = 2^{-1}\omega \cdot (1 + \sqrt{-1}) \cdot (a + b\sqrt{-1})/(a - b\sqrt{-1}).$$

We can see that $N_{E_v/F_\mathfrak{p}}(\epsilon) = \eta$ and

$$\begin{aligned}
\epsilon &= 2^{-1}\pi_F^{-3}\omega \cdot (1 + \sqrt{-1}) \cdot (a^2 - b^2 + 2ab\sqrt{-1}) \\
&= 2^{-1}\pi_F^{-3}\omega \cdot \big[(a^2 - b^2 - 2ab) + (a^2 - b^2 + 2ab)\sqrt{-1}\big] \\
&= \big(\omega/2 - b\omega(a + b)/\pi_F^3\big) + \big(\omega/2 - b\omega(b - a)/\pi_F^3\big)\sqrt{-1}.
\end{aligned}$$

In the following we will prove $a \not\equiv \pm b \bmod 2$. If not then $a = \pm b + 2u$ for some $u \in \mathfrak{o}_{F_\mathfrak{p}}$. We have $(\pm b + 2u)^2 + b^2 = \pi_F^3$. Note that $b \in \mathfrak{o}_{F_\mathfrak{p}}^\times$, then

$$2 = \mathrm{ord}_\mathfrak{p}(4u^2 + 2b^2 \pm 4ub) = \mathrm{ord}_\mathfrak{p}(\pi_F^3) = 3,$$

a contradiction. So $a \not\equiv b \bmod 2$. Since a, b are units and the residue field of $\mathfrak{o}_{F_\mathfrak{p}}$ is $\mathbb{F}_2$, we have

$$a + b = \pi_F u_1, \quad a - b = \pi_F u_2$$

with $u_1, u_2 \in \mathfrak{o}_{F_\mathfrak{p}}^\times$. Then

$$\begin{aligned}
\epsilon &= \big(\omega/2 - bu_1\omega/\pi_F^2\big) + \sqrt{-1}\big(\omega/2 + bu_2\omega/\pi_F^2\big) \\
&= 2^{-1}\omega\big(1 - bu_1 \cdot (2/\pi_F^2)\big) + 2^{-1}\omega\big(1 + bu_2 \cdot (2/\pi_F^2)\big)\sqrt{-1}.
\end{aligned}$$

We have $\epsilon \in \mathfrak{o}_{F_\mathfrak{p}} + \mathfrak{o}_{F_\mathfrak{p}}\sqrt{-1}$ since $b, u_1, u_2 \in \mathfrak{o}_{F_\mathfrak{p}}^\times$ and the residue field of $\mathfrak{o}_{F_\mathfrak{p}}$ is $\mathbb{F}_2$.

Let $\mathfrak{q}$ be the unique prime of $\mathbb{Q}(\sqrt{-1})$ above 2. Let $s \in \mathbb{Z}_2^\times$ such that $s^2 = p/3$ or $-p/3$. For any $\delta \in E_v^*$ that satisfies $N_{E_v/F_\mathfrak{p}}(\delta) = \eta$, we have

$$\begin{aligned}\left(\frac{\delta, p}{v}\right)_4 &= \left(\frac{\delta, 3}{v}\right)_4 \cdot \left(\frac{\delta, s}{v}\right) \cdot \left(\frac{\delta, \pm 1}{v}\right)_4 \\ &= \left(\frac{\delta, 3}{v}\right)_4 \cdot \left(\frac{\eta, s}{\mathfrak{p}}\right) \cdot 1 = \left(\frac{\delta, 3}{v}\right)_4 \cdot \left(\frac{1, s}{p}\right) = \left(\frac{\delta, 3}{v}\right)_4.\end{aligned}$$

Then

$$\begin{aligned}\left(\frac{\epsilon, p}{v}\right)_4 &= \left(\frac{2^{-1}\pi_F^{-3}\omega \cdot (1+\sqrt{-1}) \cdot (a+b\sqrt{-1})^2, 3}{v}\right)_4 \\ &= \left(\frac{2^{-1}, 3}{v}\right)_4 \cdot \left(\frac{\pi_F^3, 3}{v}\right)_4^{-1} \cdot \left(\frac{\omega, 3}{v}\right)_4 \cdot \left(\frac{1+\sqrt{-1}, 3}{v}\right)_4 \cdot \left(\frac{a+b\sqrt{-1}, 3}{v}\right) \\ &= \left(\frac{2^{-1}, 3}{\mathfrak{q}}\right) \cdot \left(\frac{\pi_F^3, 3}{v}\right)_4^{-1} \cdot \left(\frac{\pm 2, 3}{\mathfrak{q}}\right)_4 \cdot \left(\frac{1+\sqrt{-1}, 3}{\mathfrak{q}}\right) \cdot \left(\frac{\pi_F^3, 3}{\mathfrak{p}}\right) \\ &= 1 \cdot \left(\frac{\pi_F^3, 3}{v}\right)_4^{-1} \cdot 1 \cdot (-1) \cdot \left(\frac{\pi_F, 3}{\mathfrak{p}}\right).\end{aligned}$$

Since $\left(\frac{\pi_F, -1}{\mathfrak{p}}\right) = 1$ by the choice of π_F and since $F_\mathfrak{p}(\sqrt{-3})/F_\mathfrak{p}$ is unramified of degree 2, one obtains

$$\left(\frac{\pi_F, 3}{\mathfrak{p}}\right) = \left(\frac{\pi_F, -3}{\mathfrak{p}}\right) = -1.$$

Therefore

$$\left(\frac{\epsilon, p}{v}\right)_4 = \left(\frac{\pi_F^3, 3}{v}\right)_4^{-1} = \left(\frac{N_{E_v/\mathbb{Q}_2}(a+b\sqrt{-1}), 3}{\mathfrak{q}}\right)_4^{-1}.$$

We can write $N_{E_v/\mathbb{Q}_2}(a+b\sqrt{-1}) = 2^3 m$ with $m \equiv 1$ or $-3 \bmod 8$. Therefore

$$\left(\frac{\epsilon, p}{v}\right)_4 = \left(\frac{2^3 m, 3}{\mathfrak{q}}\right)_4^{-1} = \left(\frac{m, 3}{\mathfrak{q}}\right)_4^{-1} = \begin{cases} 1 & \text{if } m \equiv 1 \bmod 8, \\ \left(\frac{-3,3}{\mathfrak{q}}\right)_4^{-1} = 1 & \text{if } m \equiv -3 \bmod 8. \end{cases}$$

Let $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \varepsilon_F$. Since $\eta = \varepsilon_F^{i_0}$, we have

$$\left(\frac{\epsilon, p}{v}\right)_4 = \left(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\right)_4^{i_0}$$

by Lemma 1.6. Since i_0 is odd by Lemma 2.1, one obtains

$$\left(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, p}{v}\right)_4 = 1. \qquad \square$$

Corollary 2.3. *Let $F = \mathbb{Q}(\sqrt{2d})$ and $p|d$, $p \equiv \pm 3 \bmod 8$. If one of the equations $x^2 - 2dy^2 = \pm 2$ is solvable over $\mathbb{Z}$, then $x^2 + y^2 = \varepsilon_F$ is not solvable over $\mathfrak{o}_F$.*

Proof. Recall $\mathbf{X}_{\varepsilon_F}$ is the affine scheme over $\mathfrak{o}_F$ defined by $x^2 + y^2 = \varepsilon_F$. Let $\Theta = E(\sqrt[4]{p})$. One obtains

$$\psi_{\Theta/E}\Big(f_E\Big[\prod_{\mathfrak{p}\le\infty}(x_\mathfrak{p}, y_\mathfrak{p})\Big]\Big) = \Big(\frac{x_{\mathfrak{p}'} + y_{\mathfrak{p}'}\sqrt{-1}, p}{v}\Big)_4 = -1$$

for any $\prod_{\mathfrak{p}\le\infty}(x_\mathfrak{p}, y_\mathfrak{p}) \in \prod_{\mathfrak{p}\le\infty}\mathbf{X}_{\varepsilon_F}(\mathfrak{o}_{F_\mathfrak{p}})$ by Lemmas 2.1 and 2.2, where $\psi_{\Theta/E}$ is the Artin map, v and $\mathfrak{p}'$ are respectively the unique prime of E and F above p. The result follows from the class field theory. □

Remark. In fact $x^2 + y^2 = \varepsilon_F$ has local integral solutions at every place of F. The solvability is obvious if the place is not above 2. Let $\mathfrak{p}$ be the unique place of F above 2. Then we only need to show that it is solvable over $\mathfrak{o}_{F_\mathfrak{p}}$. Since $\eta = \omega^2/2 = \varepsilon_F^{i_0}$ with i_0 odd, we have

$$x^2 + y^2 = \varepsilon_F \text{ is solvable over } \mathfrak{o}_{F_\mathfrak{p}} \iff x^2 + y^2 = \eta \text{ is solvable over } \mathfrak{o}_{F_\mathfrak{p}}.$$

Note that $\omega = x_0 + y_0\sqrt{2d}$ with $2|x_0$ and y_0 odd. It is easy to verify that $x^2 + y^2 = \eta$ is solvable over $\mathfrak{o}_{F_\mathfrak{p}}$ by [3, Theorem 1].

Theorem 2.4. *Let $F = \mathbb{Q}(\sqrt{2d})$. If $p|d$, $p \equiv \pm 3 \bmod 8$ and one of the equations $x^2 - 2dy^2 = \pm 2$ is solvable over $\mathbb{Z}$, then the Diophantine equation $x^2 + y^2 = \alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of Θ and H_L, where $\Theta = E(\sqrt[4]{p})$ and H_L is the ring class field corresponding to the order $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$.*

Proof. Let $\mathfrak{p}$ be a place of F and $L_\mathfrak{p}$ the $\mathfrak{p}$-adic completion of L inside $E_\mathfrak{p} = E \otimes_F F_\mathfrak{p}$.

Let v_1 and $\mathfrak{p}_1$ be respectively the unique prime of E and F above 2. By Lemma 1.6, one has $\big(\frac{\xi, p}{v}\big)_4 = 1$ for any $\xi \in L_{\mathfrak{p}_1}^\times$ with $N_{E_{v_1}/F_{\mathfrak{p}_1}}(\xi) = 1$. This implies that

$$\lambda_E(\mathbf{T}(\mathfrak{o}_{F_{\mathfrak{p}_1}})) \subseteq N_{\Theta_{\mathfrak{V}_1}/E_{v_1}}(\Theta_{\mathfrak{V}_1}^*),$$

where $\mathfrak{V}_1$ is the unique place of Θ above v_1.

Let v_2 and $\mathfrak{p}_2$ be respectively the unique prime of E and F above p. In much the same way as in Lemma 2.1, one has $\big(\frac{\xi, p}{v}\big)_4 = 1$ for any $\xi \in L_{\mathfrak{p}_2}^\times$ with $N_{E_{v_2}/F_{\mathfrak{p}_2}}(\xi) = 1$. This implies that

$$\lambda_E(\mathbf{T}(\mathfrak{o}_{F_{\mathfrak{p}_2}})) \subseteq N_{\Theta_{v_2}/E_{v_2}}(\Theta_{v_2}^*),$$

where $\Theta_{v_2} = \Theta \otimes_E E_{v_2}$.

Since Θ/E is unramified over all primes except v_1 and v_2, the natural group homomorphism

$$\tilde{\lambda}_E : T(\mathbb{A}_F)/T(F)\prod_{\mathfrak{p}\le\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}}) \to [\mathbb{I}_E/E^* N_{\Theta/E}(\mathbb{I}_\Theta)] \times [\mathbb{I}_E/E^*\prod_{\mathfrak{p}\le\infty} L_\mathfrak{p}^\times]$$

is well-defined. By Proposition 1.1, we only need to show $\tilde{\lambda}_E$ is injective.

Let $u \in \ker\tilde{\lambda}_E$. Then there are $\beta \in E^*$ and $i \in \prod_{\mathfrak{p}\le\infty} L_\mathfrak{p}^\times$ with $\tilde{\lambda}_E(u) = \beta i$. Therefore

$$N_{E/F}(\beta) = N_{E/F}(i)^{-1} \in F^* \cap \Big(\prod_{\mathfrak{p}\le\infty} \mathfrak{o}_{F_\mathfrak{p}}^\times\Big) = \{\pm\varepsilon_F^n \mid n \in \mathbb{Z}\}.$$

Since $N_{E/F}(\beta)$ is totally positive, we have $N_{E/F}(\beta) = \varepsilon_F^n$ and $N_{E/F}(i) = \varepsilon_F^{-n}$.

Assume that n is odd. Write $i = (i_v)_v \in \mathbb{I}_E$. Since Θ/E is unramified over all primes of E except v_1 and v_2, one has that $\psi_{\Theta/E}(i_v)$ is trivial for all primes $v \ne v_1, v_2$, where i_v is regarded as an idele whose v-component is i_v and 1 otherwise. Since $N_{E_v/F_\mathfrak{p}}(i_v) = \varepsilon_F^{-n}$, one gets

$$\psi_{\Theta/E}(\beta i) = \psi_{\Theta/E}(i) = \psi_{\Theta/E}(i_{v_1})\psi_{\Theta/E}(i_{v_2}) = 1\cdot(-1)^{-n} = -1$$

by Lemmas 2.1 and 2.2, where $\psi_{\Theta/E}$ is the Artin map. This contradicts the fact that $u \in \ker\tilde{\lambda}_E$. Therefore n is even. Let

$$\gamma = \beta\varepsilon_F^{n/2}, \quad j = i\varepsilon_F^{-n/2}.$$

Then

$$N_{E/F}(\gamma) = N_{E/F}(j) = 1 \implies \gamma \in T(F) \text{ and } j \in \prod_{\mathfrak{p}\le\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}}).$$

So $\beta i = \gamma j \in T(F)\prod_{\mathfrak{p}\le\infty}\mathbf{T}(\mathfrak{o}_{F_\mathfrak{p}})$. Then $\tilde{\lambda}_E$ is injective. □

In the following we consider a special case. Let $d = p$ be a prime with $p \equiv 3 \bmod 8$. Then the equation $x^2 - 2py^2 = -2$ is solvable over $\mathbb{Z}$ (see [15, Corollary 2]).

Lemma 2.5. *Let $F = \mathbb{Q}(\sqrt{2p})$ and $p \equiv 3 \bmod 8$. If $x_\mathfrak{p}$ and $y_\mathfrak{p}$ in $\mathfrak{o}_{F_\mathfrak{p}}$ satisfy $x_\mathfrak{p}^2 + y_\mathfrak{p}^2 = \varepsilon_F$, then*

$$\Big(\frac{x_\mathfrak{p} + y_\mathfrak{p}\sqrt{-1}, 2}{v}\Big)_4 = -1,$$

where v and $\mathfrak{p}$ are respectively the unique prime in E and F above 2.

Proof. We have

$$\begin{aligned}\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},2}{v}\Big)_4 &= \Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},2p}{v}\Big)_4\cdot\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},p}{v}\Big)_4^{-1}\\ &= \Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},\sqrt{2p}}{v}\Big)\cdot 1 = \Big(\frac{\varepsilon_F,\sqrt{2p}}{\mathfrak{p}}\Big),\end{aligned}$$

where the second equation holds by Lemma 2.2. Recall $\omega = x_0 + y_0\sqrt{2d}$ and $\eta = \omega^2/2$, here $x_0, y_0 \in \mathbb{Z}$ satisfy $x_0^2 - 2py_0^2 = -2$. Since $\eta = \varepsilon_F^{i_0}$ for some $i_0 \in \mathbb{Z}$, we have that i_0 is odd by Lemma 2.1. So we deduce that

$$\begin{aligned}\Big(\frac{x_\mathfrak{p}+y_\mathfrak{p}\sqrt{-1},2}{v}\Big)_4 &= \Big(\frac{\varepsilon_F,\sqrt{2p}}{\mathfrak{p}}\Big) = \Big(\frac{\omega^2/2,\sqrt{2p}}{\mathfrak{p}}\Big)\\ &= \Big(\frac{2,\sqrt{2p}}{\mathfrak{p}}\Big) = \Big(\frac{2,-2p}{2}\Big) = -1.\end{aligned}$$

□

By Lemmas 2.2 and 2.5 and a similar argument as in the proof of Theorem 2.4, we have the following result:

Proposition 2.6. *Let p be a prime and $F = \mathbb{Q}(\sqrt{2p})$. If $p \equiv 3 \bmod 8$, then the Diophantine equation $x^2 + y^2 = \alpha$ is solvable over $\mathfrak{o}_F$ if and only if α satisfies the Artin condition of Θ and H_L, where $\Theta = E(\sqrt[4]{2})$ and H_L is the ring class field corresponding to the order $L = \mathfrak{o}_F + \mathfrak{o}_F\sqrt{-1}$.*

Now we use Proposition 2.6 to give an explicit example.

Example 2.7. Let $F = \mathbb{Q}(\sqrt{6})$. We write $N_{F/\mathbb{Q}}(\alpha) = 2^{s_1}3^{s_2}p_1^{e_1}\cdots p_g^{e_g}$ for any $\alpha = a + b\sqrt{6}$, here $a, b \in \mathbb{Z}$. Let $D(\alpha) = \{p_1, \dots, p_g\}$. Denote

$$\begin{aligned}D_1 &= \big\{p \in D(\alpha) \mid \big(\tfrac{-1}{p}\big) = \big(\tfrac{6}{p}\big) = 1 \text{ and } \big(\tfrac{2}{p}\big) = -1\big\},\\ D_2 &= \big\{p \in D(\alpha) \mid \big(\tfrac{-1}{p}\big) = -\big(\tfrac{6}{p}\big) = 1 \text{ and } \big(\tfrac{2}{p}\big) = -1\big\},\\ D_3 &= \big\{p \in D(\alpha) \mid \big(\tfrac{-1}{p}\big) = \big(\tfrac{6}{p}\big) = 1 \text{ and } \big(\tfrac{2}{p}\big)_4 = -1\big\}.\end{aligned}$$

It is easy to see that e_i is even for $p_i \in D_2$.

Then $x^2 + y^2 = \alpha$ is solvable over $\mathfrak{o}_F$ if and only if

(1) The equation $x^2 + y^2 = \alpha$ has integral solutions at every place of F.

(2) $D_1 \neq \emptyset$, or $2|a$, or

$$\sum_{p_i\in D_2} e_i/2 + \sum_{p_i\in D_3} e_i \equiv \begin{cases}0 \bmod 2 & \text{if } a \equiv \pm 1 \bmod 8,\\ 1 \bmod 2 & \text{if } a \equiv \pm 3 \bmod 8\end{cases}$$

for $D_1 = \emptyset$ and $2 \nmid a$.

Acknowledgments. The author thanks the referee for many useful suggestions.

Bibliography

[1] J.-L. Colliot-Thélène and F. Xu, Brauer–Manin obstruction for integral points of homogeneous spaces and representations by integral quadratic forms, *Compositio Math.* **145** (2009), 309–363.

[2] D. Harari, Le défaut d'approximation forte pour les groups algébriques commutatifs, *Algebra Number Theory* **2** (2008), 595–611.

[3] O. Körner, Integral representations over local fields and the number of genera of quadratic forms, *Acta Arith.* **24** (1973), 301–311.

[4] J. S. Milne, *Algebraic Geometry*, World Scientific, Singapore, 1998.

[5] T. Nagell, On the representations of integers as the sum of two integral squares in algebraic, mainly quadratic fields, *Nova Acta Soc. Sci. Upsal. (4)* **15** (1953), no. 11.

[6] T. Nagell, On the sum of two integral squares in certain quadratic fields, *Ark. Mat.* **4** (1961), 267–286.

[7] I. Niven, Integers of quadratic fields as sums of squares, *Trans. Amer. Math. Soc.* **48** (1940), 405–417.

[8] O. Perron, *Die Lehre von den Kettenbrüchen*, Chelsea Publishing, New York, 1929.

[9] M. Peters, Die Stufe von Ordnungen ganzer Zahlen in algebraischen Zahlkörpern, *Math. Ann.* **195** (1994), 309–314.

[10] D. Wei, On the sum of two squares in quadratic fields $\mathbb{Q}(\sqrt{\pm p})$, *Acta Arith.* **147** (2011), 253–260.

[11] D. Wei, Integral points for groups of multiplicative type, *Adv. Math.* **232** (2013), 36–56.

[12] D. Wei, On the Diophantine equation $x^2 - Dy^2 = n$, *Sci. China Math.* **56** (2013), 227–238.

[13] D. Wei, On the sum of two integral squares in the imaginary quadratic field $\mathbb{Q}(\sqrt{-2p})$, *Sci. China Math.*, to appear.

[14] D. Wei and F. Xu, Integral points for multi-norm tori, *Proc. London Math. Soc.* **104** (2012), 1019–1044.

[15] H. Yokoi, Solvability of Diophantine equation $x^2 - Dy^2 = \pm 2$ and new invariants for real quadratic fields, *Nagoya Math. J.* **134** (1994), 137–149.

Received November 14, 2012; revised April 24, 2013.

Author information

Dasheng Wei, Academy of Mathematics and System Science, CAS,
Beijing 100190, P. R. China.
E-mail: dshwei@amss.ac.cn